

# Addressing Negative Commons Governance with Positive Commons Principles

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## Abstract

To evaluate whether Ostrom's eight principles for positive commons governance contribute to successful management of negative common pool resources, we perform a literature review of the current state of two different governance systems. E-waste governance and the Linux open source community have different, yet related overall goals. The management of e-waste is directly tied to improving and extending the life-cycle of physical materials. The Linux kernel on the other hand, is a form of digital infrastructure which aims to extend the software support life-cycle for various kinds of hardware. Nonetheless, they are two examples of negative commons situations – e-waste governance systems seek to reduce the amount of mismanaged waste, while the Linux community seeks to reduce the number of error-prone or malicious contributions which make their way into the main branch. Both communities have seen variable amounts of success; the Linux kernel is one of the most highly trusted and versatile pieces of software, while the governance of e-waste has historically fallen under scrutiny. We conclude that the same eight principles by Ostrom that aid positive commons governance, tend to appear in successful negative commons governance systems. Future governance designs should seek to find ways to incorporate those principles, specifically improving the boundaries regarding who can contribute to a negative pool and strengthening the overall nested structure.

## Keywords

Negative common pool resources, Linux kernel, E-waste, Basel Convention, Governance

## 1 Introduction

The tragedy of the commons, formalized in 1968 by ecologist Garret Hardin in an essay of the same name [19], concerns itself with managing human population growth. Hardin's argument leans on the observed over-exploitation of resources which are held in common, in other words, resources in arenas where it is difficult – if not impossible – to restrict public access to them. More generally, the tragedy relates to the managerial pitfall of any publicly held stock when appropriator's self-interested decisions are used to control the overall exploitation of the resource. He details the over-grazing of public land as an example. It's necessary here to draw the distinction between an individual incentive and a shared one – the owner of a herd has an individual incentive to have as many cattle as possible graze on the field, which is in tension with the

shared incentive of all appropriators to maintain the land for future grazing.

In the same essay, Hardin describes a different kind of commons: pollution. In this situation, donors (an analogue to appropriators) have an individual incentive to dump as much waste into a river as possible due to the high cost (monetary, temporal, etc) of proper disposal. In tension with the individual incentive, donors have a shared incentive to keep the river clean as to not suffer the detrimental environmental effects. Hardin notes the inverse relationship between the first problem and the second – while the appropriators of the grazing land have a shared incentive to maintain or even increase the stock, the appropriators of the polluted river have a shared incentive to keep the stock low, or eliminate it entirely. These kinds of situations, where common stock is sought to be reduced, have been dubbed *negative commons resource problems* [28, 29].

Hardin concludes that the tragedy cannot be resolved through technological innovation, and therefore addressing over-consumption requires imposing a restriction on public access (through privatized control).

Ostrom continued the analysis of CPRs, ultimately receiving the 2009 Nobel Prize in Economics for her work. Perhaps best summarized in her book *Governing the Commons* [35], she analyzes different successfully managed CPRs around the world. She compares CPRs in Japan, Switzerland, the Philippines, Spain, Sri Lanka, Türkiye, and many other communities, deriving a list of eight key principles (further described in §2.1) that lead to successful communal management of the pool.

CPRs often do appear in computing, usually manifesting themselves as technical problems like efficient job scheduling and resource allocation. Negative CPRs (NCPRs) also appear within the space; however, they tend to arise from institutional challenges rather than purely technical constraints – for example, the accumulation of bugs within a code base or the generation of e-waste. Taking inspiration from Ostrom's own work, we seek to analyze examples of governance structures around these NCPRs. Through qualitative analysis of the aforementioned NCPRs, we find that successful managerial systems tend to still utilize Ostrom's eight principles.

To further formalize the problem, we seek to answer:

- To what extent does the institutional design of an NCPR governance structure adhere to Ostrom's eight design principles for CPRs?
- To what extent do local manifestations of the NCPR governance structure adhere to Ostrom's eight design principles for CPRs?

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- Broadly, do Ostrom’s eight principles contribute to successful management of NCPRs?

## 2 Background

We begin by giving a description of the taxonomies and classifications used throughout this paper, followed by a brief description of the two NCPR governance structures we analyze.

### 2.1 Ostrom’s Eight Principles

Ostrom’s pivotal work, *Governing the Commons* [35], performs a thorough analysis of successful governance structures of CPRs around the world. She distills out the following eight guiding principles:

- (1) **Clearly defined boundaries:** The extents for who can appropriate from – or contribute to – a CPR must be well determined. This includes clear definitions of the resource itself, its boundaries, and the conditions under which participation is permitted.
- (2) **Congruence between appropriation/provision rules and local conditions:** Rules related to how much or how often a resource is extracted from the pool must be related to the local conditions of the pool itself.
- (3) **Collective-choice arrangements:** Rules governing the commons must be set by those who the rules affect.
- (4) **Monitoring:** Appropriator behaviors and the state of the resource pool must be monitored.
- (5) **Graduated sanctions:** Violations of rules must be met with sanctions, the seriousness of which must depend on the seriousness of the infraction.
- (6) **Conflict-resolution mechanisms:** An accessible method for resolving issues between appropriators, monitors, or otherwise any participants in the commons must exist.
- (7) **Minimal recognition of rights to organize:** Appropriators must be able to create their own governance structures; these local structures must be recognized by all higher-level governance structures.
- (8) **Nested enterprises:** CPRs that are part of larger systems must be organized into multiple layers. This goes hand in hand with the right of appropriators to organize.

### 2.2 Negative Commons Taxonomy

Compared to positive commons governance, significantly less literature exists regarding how to properly manage negative commons situations. Martin Deron, in a 2025 LIMITS paper [14], outlines a useful taxonomy of NCPRs. Deron was primarily focused on understanding negative commons situations in the context of a societal transition towards a more sustainable future. Deron’s contribution is heavily based on Monnin’s prior work with negative commons [30, 31], in which he creates four categories of societal aspects that either linger and slow the transition towards sustainability or enable that transition to take place:

- (1) **Ruins:** Material elements (e.g. network infrastructure, excavation sites) which have outlived their usefulness, yet remain standing.

- (2) **Ghosts:** Immaterial, ideological elements which paved the way for unsustainable ruins to form, including paradigms, expectations, and beliefs.
- (3) **Seeds:** Material elements which can define how ICT technology could work in a more sustainable future (e.g. localized mesh networks, community-based repair cafes).
- (4) **Visions:** Immaterial elements which support the emergence of seeds in an alternative, sustainable future.

We further distill this taxonomy into just two categories, immaterial and material NCPRs, and consider a case study for each type. The goal is to determine whether the governance structure is able to address the needs of the NCPR, and then compare and contrast how well Ostrom’s principles apply to both.

### 2.3 E-waste

E-waste, or Waste Electrical and Electronic Equipment (WEEE), is recognized as one of the fastest-growing solid waste streams in the world. In 2022 alone, an estimated 62 million tons of e-waste were produced globally, a figure projected to rise significantly in the coming decade [8]. This waste stream is physically complex and comprises over 60 different elements from the periodic table. It contains both valuable resources, including gold, copper, silver, and critical raw materials like cobalt and indium, alongside a cocktail of hazardous materials such as lead and mercury.

The processing of this waste often occurs outside of formal industrial facilities. A significant portion of global e-waste is handled through “informal recycling”, where rudimentary methods are used to extract valuable metals. These methods include open burning of electrical cables to recover copper, and acid leaching to extract gold from circuit boards [8]. These processing practices release heavy metals and organic pollutants directly into the surrounding environment (soil, air, water), exposing workers and entire communities to severe health risks.

**2.3.1 E-waste and Negative Commons.** E-waste is a clear example of material NCPR. As with other pollution-based negative commons problems, actors have incentives to externalize disposal costs due to the high cost of proper processing, and the availability of informal dumping or low-quality recycling. This leads appropriators to generate e-waste while overlooking the environmental and health externalities. In this context, the goal of the governance structure is to reduce the stock of e-waste and ensure that the costs of proper disposal are borne by the actors who generate it, rather than by downstream communities and environments.

**2.3.2 Global E-waste Trade and the Basel Convention.** While e-waste is generated globally, its movement follows a distinct geographic pattern, typically flowing from the industrial North to the Global South. This flow is driven primarily by economic disparities; the cost of extracting materials safely in developed nations is often higher than the value of the recovered materials. Conversely, in developing nations, low labor costs make labor-intensive manual dismantling profitable [27, 39].

To combat the trans-boundary movement of e-waste and other hazardous waste, the Basel Convention was established in 1989 [17] (§4.1). It restricts such movements and has since been ratified

by 191 nations, making it one of the most influential international policies governing global e-waste flows.

**2.3.3 Local E-waste Governance.** On the state or local level, many countries have established their own management systems for e-waste, including the EU's WEEE Directive [10], the USA's state-level frameworks (e.g., California's SB 20/50) [37], and Brazil's National Solid Waste Policy (PNRS) [36]. While other nations maintain some degree of e-waste management, there is no universal baseline. Generally, Global North countries possess more developed regulatory frameworks than the Global South [27].

## 2.4 Open Source Software and Linux

Free and open source software (FOSS) is foundational for the vast majority of systems we use today [20]. The Linux operating system is perhaps one of the most famous examples of such, created in 1991 by Linus Torvalds [9]. The Linux Foundation [1] reports that at the time of writing, there are 6,773 unique annual contributors creating commits, pull requests, and raising issues [16].

Linus Torvalds' own role in this community has been studied extensively; he acts as a leader, an arbiter, a mediator, but also an active contributor in technical discussions [34, 41]. All communication and coordination is done via mailing lists.

Patches are sent by individual contributors to Maintainers, members of the Linux community who are in charge of a particular subsystem. Maintainers are selected by the community; they tend to be people who have working experience with that particular subsystem. Linux goes through a release cycle roughly every 2-3 months, during which Torvalds selects patches sent to him by Maintainers to be included in the next version [4]. This chain-of-trust allows the community to function even with the large size of the project – by the time a patch makes it to Torvalds, it's generally believed to be a good addition to the kernel [4].

**2.4.1 Linux and Negative Commons.** Contributing to the kernel carries with it positive social status within the broad software development community. As a result, there is an individual incentive to get as many changes as possible merged into the main branch. This individual incentive is in tension with the shared incentive to maintain high code quality. A positive commons lens on this situation would then result in claiming that “high code quality” is the stock being managed by the community. However, this lens falls short – contributors are not extracting any particular resource from the code base, they are inherently *adding* to the system. A negative commons lens is more directly applicable to this situation, in the same way that donors add pollutants to the environment, contributors add code issues. A governance structure embedded into the community is tasked with reducing the number of bugs introduced into the code base. The structure does restrict who can contribute, but even positive commons governance structures impose limits on who can appropriate from a particular pool. The open source nature of the project is ultimately what makes a commons lens applicable. For this reason, we claim that the Linux kernel is an example of an immaterial NCPR.

**2.4.2 Bad Faith Bans.** In 2021, all contributors who used a University of Minnesota (UMN) email address were banned from submitting patches into the kernel [33]. A group of researchers were found

to be intentionally contributing bugs. In response, Linux leadership decided to retroactively revert all patches previously submitted from UMN [18].

This is one of the largest scale bans which has ever taken place in the Linux community, although smaller bans due to “bad faith” submissions happen much more frequently. Oftentimes, these individuals are banned for repeatedly not listening to Maintainer feedback.

**2.4.3 Toxicity and Bugs.** Ferreira et al. [15] conduct a qualitative analysis on 1,545 emails regarding rejected changes – they find that instances of incivility appear in approximately two-thirds of the emails. The 2007 Kernel Summit [11] emphasized the importance for Maintainers to set better examples in mailing lists. Torvalds' own stance on the toxic nature of the community is maybe best exemplified in a 2013 exchange regarding patches submitted late during a release cycle [23]:

Greg, the reason you get a lot of stable patches seems to be that you make it easy to act as a door-mat ... You may need to learn to shout at people.

Another contributor, Sarah Sharp, expressed concern about these comments [38]:

Is this what we need in order to get improve *-stable*?  
Linus Torvalds is advocating for physical intimidation and violence.

In response, Torvalds continued to justify the use of harsh language, crediting it as a means of clear communication when it comes to important design decisions:

I can't just say "please don't do that", because people won't listen. I say "On the internet, nobody can hear you being subtle", and I mean it.

Leadership views *harsh* communication as being a part of *honest* communication, and therefore a means of keeping the kernel bug free. This exchange, alongside plenty of others [12], ultimately led to the adoption of the Contributor Covenant Code of Conduct (CCCC) [2], and the creation of the Code of Conduct Committee.

## 3 Methods

We use the governance structures around e-waste and the Linux kernel as two case studies of NCPR management systems found within computing. To evaluate the applicability of positive commons design principles to these NCPRs, we employ a qualitative, comparative, case study approach. First, we perform a survey of primary sources to determine how the governance structure attempts to manage stock. We then perform a qualitative coding exercise to identify whether or not there exists sufficient evidence of utilizing Ostrom's eight principles for CPRs in each NCPR. Finally, we evaluate whether or not each NCPR is considered successful in managing its respective stock, and observe whether or not there exists a correlation with the identified positive commons principles.

### 3.1 Data Collection

Data was derived from a survey of existing literature, including academic analysis of governance structures, technical field reports, email communications, and policy frameworks.

| Region(s)      | Number of papers |
|----------------|------------------|
| United States  | 1                |
| Brazil         | 2                |
| Netherlands    | 1                |
| South Africa   | 1                |
| Ghana          | 1                |
| China          | 1                |
| Multi-regional | 3                |

**Table 1: Geographic distribution of the WEEE-related papers reviewed.**

**3.1.1 E-waste Data.** To investigate the governance of electronic waste through an NCPR lens, we conducted a comprehensive literature review focusing on real-world policy implementations and regulatory frameworks.

We performed a systematic search using Google Scholar to identify relevant academic literature. The search query utilized a combination of the following keywords: “E-Waste”, “WEEE”, “governance”, “commons”, and “Ostrom”. This specific combination was chosen to bridge the gap between general waste management literature and theoretical frameworks of polycentric governance and common-pool resources. The initial search yielded approximately 625 results.

We filter down the documents to ones that analyzed specific, real-world case studies of e-waste governance and explicitly discussed regulatory policies or institutional arrangements. Consequently, purely theoretical papers lacking geographical application, as well as articles focused solely on technological innovation or quantitative waste metrics without policy analysis, were excluded.

From the initial pool, we narrowed the selection down to a final corpus of 10 articles that best exemplified diverse governance approaches. We ensured the selection represented a global cross-section of economic development stages and regulatory environments. Table 1 shows the geographic distribution of the selected case studies.

**3.1.2 Linux Data.** Most Linux communication is fully public, done via the Linux Kernel Mailing List (LKML). This archive provides a very rich record of the communication between different members that resulted in particular policy or contribution process change, as well as the feedback that Maintainers provide for particular patches.

To better understand the actual day-to-day manifestations of particular policies and attitudes, we construct a dataset of LKML emails. To narrow down the number of emails considered, we queried LKML for appearances of “Nacked-by:” tags (which appear within emails where Maintainers reject a particular patch), and chose to analyze the 60 most recent, unique threads. Manual inspection reveals that seven of those threads were discussing closed patches which were rejected in the past. We remove these threads from our final dataset, keeping only the threads which concern themselves with active maintenance of the kernel. The results of the coding exercise are shown in table 2.

| Code                           | Occurrences |
|--------------------------------|-------------|
| Suggested Fix                  | 22          |
| Big Picture                    | 20          |
| Bad Style                      | 14          |
| Logical Issue                  | 13          |
| Contested (Maintained)         | 13          |
| Bad Faith                      | 12          |
| Long-term Concern              | 8           |
| “Keep the NAK”                 | 8           |
| Uncivil/Rude                   | 5           |
| Known Issue in Other Subsystem | 5           |
| Bad Patch Format               | 5           |
| Shutdown                       | 5           |
| Contested (Overturned)         | 4           |
| Multiple Maintainers Criticize | 4           |
| “Kernel Dev Identity”          | 3           |
| Cite Rules                     | 2           |
| Slippery Slope                 | 1           |
| No Tests                       | 1           |

**Table 2: Codes identified in evaluating LKML emails.**

## 4 Evaluating Ostrom’s Principles

### 4.1 The E-waste NCPR

In this section, we evaluate whether the e-waste NCPR aligns with each of Ostrom’s eight principles.

**4.1.1 Clearly defined boundaries.** The Basel Convention attempts to harden the boundaries through Article 1 and the 2022 “E-waste Amendments” (entries A1181 and Y49), which legally subject all transboundary e-waste to Prior Informed Consent (PIC) [17]. However, this legal boundary frequently dissolves in practice due to the ambiguity between “waste” and “used goods”. While the European Union reinforces Basel’s boundary by requiring proof of functionality for exports, the Global South faces a “porous border” reality [10, 39]. In the West Bank, geopolitical fragmentation creates a governance vacuum where e-waste hubs emerge simply because the territory lies outside effective national jurisdiction [13]. Similarly, in the U.S. (a non-party of the Basel Convention), market actors exploit these definitional gaps to classify exports as “digital bridge” philanthropy, effectively smuggling waste through the “repair loop-hole” [37]. Thus, while Basel provides a de jure boundary, the de facto boundary is fluid and contested.

**4.1.2 Congruence Between Appropriation Rules and Local Conditions.** The Basel Convention’s primary appropriation mechanism assumes a level of bureaucratic capacity that is incongruent with local realities. As a result, the convention lacks any meaningful provisions for local adaptation among its parties. A primary example is the requirement for “Environmentally Sound Management” (ESM) based on “Best Available Techniques” (BAT) [17]. While establishing high standards, this mandate fails to account for global technical and economic discrepancies. In China and Ghana, both ratified states of Basel Convention, top-down attempts to impose this formal, high-cost model (e.g., via “eco-parks”) failed because they

ignored the low operating costs and high recovery skills of the informal sector, driving the trade back underground [7, 24]. Conversely, small-scale "constructed governance" in Brazil did demonstrate high congruence. Instead of relying on distant bureaucratic forms, rules regarding pricing and logistics were negotiated directly between cooperatives and private buyers, resulting in higher compliance than international mandates could achieve [36].

**4.1.3 Collective Decision-making.** The Basel Convention limits the participation to the state level via the Conference of the Parties (COP) [17], effectively excluding the informal workers who manage the bulk of global flows. This exclusion is mirrored in South Africa, where the "WEEE Producer Forum" marginalized key stakeholders to protect industry interests, leading to legitimacy crises and paralysis [22]. In contrast, we found that effective collective decision-making is observed where governance is localized. The Netherlands utilizes a "monitoring council" where producers and municipalities jointly evaluate the system, allowing for adaptive management [10].

**4.1.4 Monitoring.** The Basel Convention relies on Article 13 (national reporting), a "fire-alarm" system that is functionally broken, with over 50% of parties failing to report harmonized statistics in 2013 [17, 32]. This failure creates a global monitoring vacuum, filled in the U.S. by a patchwork of competing market certifications (R2 vs. e-Stewards) that offer inconsistent transparency [37]. Even though substantial effort has been devoted to monitoring e-waste, including institutionalized and socialized monitoring mechanisms, significant challenges remain. For example, the Netherlands maintains a centralized national registry that aggregates data across the entire e-waste chain [10]. However, the particular characteristics of e-waste mean that data quality still suffers from major inaccuracies, which are compounded by the blurry boundaries described in subsection 4.1.1.

**4.1.5 Graduated Sanctions.** The Basel Convention's primary sanction (Article 9) is a binary instrument (Legal vs. Illegal) that lacks an enforcement arm [17], rendering it ineffective in contested zones like the West Bank where burning of e-waste persists without penalization [13].

**4.1.6 Conflict-Resolution Mechanisms.** The Basel Convention's Implementation and Compliance Committee (ICC) offers a facilitation body, but it is designed for diplomatic disputes between states, not for the daily operational conflicts of the waste trade. Without accessible mechanisms at the international level, disagreements within local governance structures tend to produce fragmentation rather than resolution. In South Africa, the original multi-stakeholder e-waste body (eWASA) was intended to serve as a unified governance arena. However, distrust in the organization's leadership and participatory processes led a group of manufacturers to form a rival body, the SA WEEE Producers' Forum [22]. Rather than resolving the underlying disagreement about collective versus competitive recycling strategies, the split divided resources, created confusion among participants, and delayed the governance transition. A similar pattern emerged in the U.S., where disagreements during the EPA's development of the R2 recycling standard led BAN to withdraw and create the rival e-Stewards certification [37]. The two standards promote incompatible approaches to e-waste exports, yet

no shared mechanism exists to adjudicate between them. In both cases, the absence of a trusted conflict-resolution process caused e-waste governance to splinter.

**4.1.7 Minimal Recognition of Rights to Organize.** The Basel Convention remains largely silent on informal sector rights, treating e-waste management as a technical state function [27, 39]. This silence often enables national governments in places like Ghana and China to criminalize or displace self-organized scrap hubs under the guise of "modernization" or Nature-based Solutions (NbS), explicitly violating the right to organize [24, 39]. This criminalization drives the trade underground, stripping it of oversight and paradoxically fueling a surge in unregulated, illicit activity.

**4.1.8 Nested Enterprises.** The breakdown of the nested network is a primary structural failure of the global e-waste landscape under the institutional governance of Basel Convention. While the Basel Convention attempts to link international obligations to national action via Regional Centres (BCRCs), the chain frequently breaks at the national level [17, 27]. In China, central mandates decouple from local realities because local governments lack the incentive to enforce them against a profitable informal sector [24]. Conversely, the Dutch model exemplifies successful nesting on a continental scale. The European WEEE Directive (constitutional level) is tightly coupled with national law (collective-choice level) and municipal execution (operational level), supported by robust information exchange [10]. This contrast suggests that the Basel Convention's design is theoretically nested but practically disconnected in regions lacking strong institutional transmission belts.

## 4.2 The Linux NCPR

In this section, we evaluate whether the Linux NCPR aligns with each of Ostrom's eight principles.

**4.2.1 Clearly defined boundaries.** The hierarchical organization of the kernel into subsystems defines boundaries within which Maintainers exert control. Further, the patch review process itself acts to clearly define who is able to make a contribution into the code base. This principle heavily aligns itself with the *Bad Faith* code, denoting times Maintainers rejected patches due to the person making the contribution. To give an example [40], the following was coded as *Bad Faith*:

The request\_irq() change in ath\_ahb\_probe() is literally the same change that you sent once and that we had to revert. Not taking any more of these, sorry.

Given the supporting institutional structure and the prevalence of these codes, we conclude that there is more than sufficient evidence of clearly defined boundaries.

**4.2.2 Rule congruence to local conditions.** Maintainers often set their own standards for what contributions are acceptable within each subsystem, as defined in the CCCC. Highly critical systems often move slower, with stricter requirements for incremental changes. The *Big Picture* code most directly aligned with this principle, representing times a Maintainer rejected patch due to it containing features which were out of scope for a particular subsystem. For example [21]:

I don't understand why patch 1 is a step in that direction, and you seem incapable of explaining it. So please either follow my suggestion on how to proceed with patch 2 without patch 1 in current form.

Again, we find sufficient evidence for this principle.

**4.2.3 Collective-choice arrangements.** Maintainers are chosen by the community [3], which determines how subsystem-level rules are created and enforced. Beyond that, the community has been able to form institution-level governance bodies, for instance, the CCCC. The *Contested (Overturned)* code also contributes to evidence for collective-choice arrangements, denoting times a Maintainer originally rejected a patch but was then convinced by the contributor to include it. Further, the *"Keep the NAK"* code denotes times a Maintainer specifically said to include their rejection of the patch if it is ever resubmitted, so they can get a chance to voice their opinion in the future decision. Once more, we conclude that there is good evidence for this principle.

**4.2.4 Monitoring.** Maintainers are monitors. Maintainers must read every submitted patch, and no patch ever makes it into the main branch without their approval. Given the exhaustive nature of this requirement, monitoring is one of the central premises used by the Linux NCPR.

**4.2.5 Graduated sanctions.** The most frequent code, *Suggested Fix*, denotes times a patch was rejected, however the Maintainer recommended a fix for the contributor to implement to get it approved. Most times, there is no consequence to submitting a bug. However if a pattern of *Bad Faith* submissions is found, the individual could be banned from further contributing to a subsystem. As an example [25]:

You are purposefully engaging in malicious compliance here...I suggest you find another part of the kernel to work upon.

In the most extreme cases (such as the UMN affair), entire organizations have been banned from contributing to any subsystem. The level of penalty depends upon the severity of the infraction, and therefore, we claim that this NCPR makes heavy use of graduated sanctions.

**4.2.6 Conflict-resolution mechanisms.** The CCCC established the Code of Conduct Committee to resolve non-trivial problems, as they come up. The rules of the CCCC are sometimes cited in emails (coded as *Cite Rules*), to direct the conflict resolution process in a productive direction. [26] is an example of such, where a patch was improperly formatted, and after a terse, tense, exchange, the rules were invoked to settle the matter. Due to the clear definitions of responsibilities, alongside the invocations of the rules themselves, we conclude that this principle is also central to the success of the governance structure.

**4.2.7 Recognition of the right to organize.** Linux subsystems have been further split into sub-subsystems, with their own Maintainers. The hierarchical nature of the chain of trust, alongside its ability to be extended, then becomes concrete evidence for the right to organize. Further, the adoption of the CCCC was a bottom-up process, providing evidence for institutional change spearheaded through the organization of individual contributors.

**4.2.8 Nested enterprises.** When Maintainers accept patches, they are further reviewed by a "higher-level" organization of Maintainers, up the chain-of-trust. Torvald's can reject patches submitted by any Maintainer. The overall structure of this system lends itself to the definition of a nested enterprise.

## 5 Discussion

The Linux community is able to sustain a kernel which powers the top 500 supercomputers [5] and approximately 63% of servers worldwide [6]. It is a highly trusted and robust system; the community demonstrates strong evidence of all eight principles in action. Conversely, the governance of e-waste, particularly under the Basel Convention, shows significant gaps, specifically in the application of effective monitoring, graduated sanctions, and nested enterprises.

### 5.1 The Importance of Boundaries and Sanctions

A critical disparity lies in the definition of boundaries. Within the Linux kernel, the boundary is a part of the fundamental infrastructure of the community. The "chain of trust" ensures no contribution enters the main branch without Maintainer approval. In contrast, e-waste governance relies on porous legal definitions of "waste" versus "used goods" – illicit flows often slip through.

Furthermore, the Linux model extensively utilizes graduated sanctions, ranging from code critiques to bans, allowing for nuanced enforcement. The Basel Convention relies on a binary legal status (legal vs. illegal) without a global enforcement arm, often resulting in no effective sanction for violations.

### 5.2 The Broken Nest

Linux succeeds through a nested structure; Maintainers connect individual contributors to top-level governance, ensuring rules align with local technical realities. E-waste governance, however, suffers from a broken nested network. International mandates often disconnect from the operational reality of the informal sector in the Global South. By failing to integrate these stakeholders, the governance lacks that congruence, paradoxically driving the trade underground.

### 5.3 Limitations and Future Work

It would be disingenuous for us to claim that the two case studies featured in this paper are perfectly analogous to one another, and that one governance system is directly applicable to the other situation. For instance, the fundamental material/immaterial difference between them presents a gap that must be bridged when attempting to compare them directly. Nonetheless, we find that in conducting an analysis of vastly different situations, it becomes much easier to ask: *why do we struggle with the management of one system, and not the other? What makes them different?* Observing these differences, in an explicit manner, allows us to then refine our understanding down to specific policies and to imagine different futures. After all, a sustainable future includes the proper management of all kinds of negative commons.

We acknowledge that this study does not measure the total quantitative effect of the Linux ecosystem nor the Basel Convention. Instead, we grounded our analysis in the presence of governing

principles at the “constitutional” level of these nested networks. We recommend the academic community to further quantify the policy hull of governance structures. This approach is expected to yield quantitative evidence to inform negative commons policy-making.

Additionally, the choice of framework informs how we evaluate each case study. Monnin [28] had previously laid out a slightly different framework than Deron’s [14], splitting negative commons into ones we must *live with*, *live with differently*, and *live without*. The two case studies in this paper fit squarely into the first two categories – however future work is necessary to analyze the third category of *live without*, to understand if CPR principles remain applicable when the goal is the total cessation of a resource’s production, rather than its management.

## 6 Conclusion

We set out to determine whether Ostrom’s design principles, originally derived for preserving positive commons, are applicable to the management of negative common pool resources. Our comparative analysis suggests a strong correlation. The successful mitigation of bugs within the Linux kernel mirrors the community’s strict adherence to all eight design principles. Conversely, the struggles of the Basel Convention illustrate the fragility of governance structures that only partially adhere to these principles while significantly neglecting others.

Empirical evidence from the local manifestations of these structures confirms that Ostrom’s framework contributes to the successful management of NCPRs. Future policy frameworks should therefore prioritize these design principles when crafting institutional designs aimed at managing negative commons, moving beyond simple mandates. Further, we urge the LIMITS community to find more examples of successful NCPR governance systems; in the comparisons of extremes, lie key insights into systemic operations.

## 7 Acknowledgments

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